

Appendix: Self-regulation of the collisionless pile-up in cooled Bondi accretion onto light PBHs

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1 Purpose of this appendix

This appendix addresses a possible failure mode of the one-dimensional Bondi+cooling calculation for very light primordial black holes (PBHs) embedded in stellar interiors. The concern is that, although the gas is collisional at the Bondi radius, the flow can become collisionless at smaller radii. Once this happens, individual particles retain their thermal transverse velocities and therefore their angular momenta. Since most of these angular momenta exceed the general-relativistic capture threshold, most particles miss the black hole on any single pass.

If the entire problem were collisionless, the accretion rate would be set by the gravitational-capture cross-section rather than by the Bondi supply. For non-relativistic particles in a Maxwellian bath,

$$\frac{\dot{M}_{\text{coll}}}{\dot{M}_B} \sim \frac{c_s^2}{c^2} \simeq 3 \times 10^{-6}, \quad (1)$$

which would catastrophically suppress the accretion rate. The central result of this appendix is that this fully collisionless limit does *not* apply over the mass range studied in the main paper. The gas is still collisional at the Bondi radius, so a hydrodynamic supply is established. The inner collisionless region modifies this supply only through a moderate pressure feedback.

For the fiducial case $M = 10^{-16} M_\odot$ in the solar core, the final hydro-kinetic closure gives

$$\chi \equiv \frac{\dot{M}}{\dot{M}_B} \simeq 0.7, \quad (2)$$

corresponding to a $\simeq 30\%$ reduction rather than the factor $\sim 10^6$ suppression in Eq. (1). The correction decreases rapidly with increasing PBH mass and is negligible by $M \gtrsim 10^{-15.5} M_\odot$.

The physical reason is a self-regulating sequence:

$$\text{missed particles} \rightarrow \text{cooling and circularization} \rightarrow \rho_{\text{pile}} \uparrow \rightarrow \lambda_{\text{mfp}}/r \downarrow \rightarrow \text{collisionalization} \rightarrow \text{angular-momentum} \quad (3)$$

Thus, in this regime, the PBH is not fed primarily by direct single-pass loss-cone capture. It is fed by a recycling/circularization/collisionalization flow whose main effect is to exert a pressure feedback on the surrounding subsonic Bondi flow.

2 Fiducial scales and ordering of radii

We consider a PBH of mass $M = 10^{-16} M_\odot$ embedded in the present solar core, with $T_\infty = 1.57 \times 10^7$ K, $\rho_\infty = 150 \text{ g cm}^{-3}$, $\mu = 0.85$, and $\gamma = 5/3$. The relevant length scales are

Bondi radius	$r_B = GM/c_{s,\infty}^2$	$\simeq 5 \times 10^{-6} \text{ cm}$
Schwarzschild radius	$r_S = 2GM/c^2$	$\simeq 3 \times 10^{-11} \text{ cm}$
Ambient Coulomb mean free path	λ_{mfp}	$\sim 10^{-7} \text{ cm}$

The gas is therefore collisional on the Bondi scale, $\lambda_{\text{mfp}} \ll r_B$. The hydrodynamic supply rate is well defined there.

In the cooled Bondi solution, however, the ratio λ_{mfp}/r increases inward. For the fiducial model, the collisionless transition occurs at

$$r_{\text{coll}} \simeq 7000 r_S \simeq 2 \times 10^{-7} \text{ cm}. \quad (4)$$

The flow at this radius is still subsonic, with $\mathcal{M} \simeq 0.7$. The cooling-induced sonic point lies deeper in, at

$$r_{\text{sonic}} \simeq 600\text{--}800 r_S. \quad (5)$$

The ordering for the fiducial case is therefore

$$r_B > r_{\text{coll}} > r_{\text{sonic}} > r_S. \quad (6)$$

This ordering is crucial. Because r_{coll} lies in the subsonic region, pressure generated by particles returning to the collisional gas can communicate upstream and modify the Bondi eigenvalue.

3 Angular momentum barrier and single-pass capture

At r_{coll} , the bulk flow is radial, but individual particles retain random transverse thermal velocities. With one-component transverse velocity dispersion

$$\sigma_\perp = \frac{c_s}{\sqrt{\gamma}}, \quad (7)$$

the characteristic specific angular momentum is

$$\ell_{\text{typ}} \sim r_{\text{coll}} \sigma_\perp \sim 30 \text{ cm}^2 \text{ s}^{-1}. \quad (8)$$

A Schwarzschild black hole captures a marginally bound non-relativistic particle only if its angular momentum is below

$$\ell_{\text{crit}} = \frac{4GM}{c} \simeq 1.8 \text{ cm}^2 \text{ s}^{-1}. \quad (9)$$

Thus

$$\frac{\ell_{\text{typ}}}{\ell_{\text{crit}}} \simeq 18. \quad (10)$$

Most particles therefore have too much angular momentum to plunge. Since the transverse velocity is two-dimensional and Rayleigh distributed, the single-pass loss-cone fraction is

$$f_{\text{cap}} \simeq \frac{1}{2} \left(\frac{\ell_{\text{crit}}}{\ell_{\text{typ}}} \right)^2 \simeq 1.5 \times 10^{-3}. \quad (11)$$

This small number is the origin of the concern. It would set the accretion rate in a fully collisionless problem. Here it does not, because the Bondi-scale gas remains collisional and missed particles can recycle or circularize rather than being permanently lost.

4 Recycling and the residence-time pile-up

Let $\dot{M}$ be the hydrodynamic mass flux supplied from the collisional region outside r_{coll} , and let f_{eff} be the effective probability per inward crossing that material is removed from the recycling loop by direct capture, angular-momentum redistribution, or collisional transport. In a steady state with such an effective sink,

$$\Phi_{\text{in}} = \frac{\dot{M}}{f_{\text{eff}}}, \quad \Phi_{\text{out}} = \frac{1 - f_{\text{eff}}}{f_{\text{eff}}} \dot{M}, \quad \dot{M}_{\text{acc}} = f_{\text{eff}} \Phi_{\text{in}} = \dot{M}. \quad (12)$$

The small loss-cone fraction therefore controls the residence time and density enhancement, not the net supplied flux itself.

The corresponding density enhancement relative to the *current* inflow is

$$\xi \equiv \frac{\Phi_{\text{in}} + \Phi_{\text{out}}}{\dot{M}} = \frac{2 - f_{\text{eff}}}{f_{\text{eff}}} \simeq \frac{2}{f_{\text{eff}}}. \quad (13)$$

If orbits were conservative and $f_{\text{eff}} = f_{\text{cap}}$, this would imply $\xi \sim 10^3$. The pile-up would be enormous. The orbits, however, are not conservative.

5 Bremsstrahlung limits the residence time

Particles lose orbital energy through bremsstrahlung during close encounters. A useful local estimate of the fractional energy loss per passage at the unperturbed density is

$$\left(\frac{\Delta E}{E} \right)_0 \simeq \frac{\varepsilon}{\rho} \frac{r}{v} \frac{1}{\frac{1}{2}v^2}, \quad (14)$$

where ε is the bremsstrahlung emissivity. In the cooled Bondi profiles this gives the representative values

r/r_S	$\mathcal{M}$	$(\Delta E/E)_0$
3000	0.86	1.1%
1000	0.96	1.3%
550	1.08	8.0%
300	1.33	21%

At an enhanced density $\xi\rho$, the cooling rate per unit mass scales as ξ . If a particle makes N passages before its orbit shrinks substantially, then

$$N \simeq \frac{1}{\xi(\Delta E/E)_0}. \quad (15)$$

Since both inbound and outbound streams contribute to the residence-time density, $\xi \simeq 2N$. Hence

$$\xi \simeq \sqrt{\frac{2}{(\Delta E/E)_0}} \simeq 5\text{--}13. \quad (16)$$

Thus the pile-up self-limits: a larger pile-up increases the cooling per particle, which shortens the residence time and lowers the pile-up.

This energy loss does *not* imply immediate plunge. To leading order, bremsstrahlung removes orbital energy more efficiently than angular momentum. Missed particles therefore shrink and circularize at approximately

$$r_{\text{circ}} \simeq \frac{\ell^2}{GM}, \quad (17)$$

rather than falling directly into the PBH. For typical particles, $\ell \sim \ell_{\text{typ}} \simeq 18\ell_{\text{crit}}$, this is $r_{\text{circ}} \sim 10^3\text{--}10^{3.5} r_S$.

6 Orbit-averaged cooling and the fate of missed particles

The single-radius estimate above is useful, but the relevant cooling is orbit-averaged and depends on angular momentum. We therefore computed $(\Delta E/E)(\ell)$ by integrating the bremsstrahlung loss along full Newtonian orbits in the converged cooled background.

The angular-momentum distribution separates into three broad populations at r_{coll} :

Population	Fraction	Angular momentum	Fate
Direct capture	$\simeq 0.2\%$	$\ell < \ell_{\text{crit}}$	swallowed
Reflected	$\simeq 33\%$	$\ell > \ell_{\text{max}}$	rethermalizes near r_{coll}
Recycling entrants	$\simeq 67\%$	$\ell_{\text{crit}} < \ell < \ell_{\text{max}}$	pile-up/circularization

For the recycling population,

$$\left\langle \frac{\Delta E}{E} \right\rangle_{\text{rec}} = 0.018, \quad \langle \xi \rangle_{\text{rec}} = 11.9. \quad (18)$$

The bulk of the pile-up is therefore controlled by high- ℓ , shallow-pericenter particles with modest cooling per orbit, $(\Delta E/E) \sim 1\text{--}2\%$, rather than by the deeply penetrating low- ℓ tail.

A Markov recycling calculation then follows individual particles through repeated passages. Because $\langle \xi \rangle \simeq 12$, the number of effective passages is $N \sim \xi/2 \sim 6$. Combining this with Eq. (11), the cumulative probability of direct GR capture before circularization is only of order one percent. Thus direct capture is not the main sink: most particles circularize. The essential question becomes whether the circularized material remains collisionless or self-collisionalizes.

7 Self-induced collisionality of the circularized pile-up

The circularized material increases the local density. Since the Coulomb mean free path scales approximately as $\lambda_{\text{mfp}} \propto \rho^{-1}$, the pile-up lowers the effective Knudsen number. At the circularization radius,

$$\text{Kn}_{\text{eff}}(\ell) = \frac{\text{Kn}_0[r_{\text{circ}}(\ell)]}{\xi(\ell)}, \quad (19)$$

where $\text{Kn}_0 = \lambda_{\text{mfp}}/r$ is evaluated from the unperturbed cooled profile. This is an orbit-averaged estimate of the local collisionality. A fully spatial calculation would compute the pile-up density by binning the orbital residence time in radius; the estimate above is sufficient for classifying which angular-momentum bins are clearly collisional, marginal, or collisionless.

The result is summarized in Fig. 2. Among the non-reflected collisionless entrants,

- $\simeq 63\%$ circularize in regions with $\text{Kn}_{\text{eff}} < 1$ and are therefore collisionalized by their own pile-up;
- $\simeq 15\%$ are marginal, with $1 < \text{Kn}_{\text{eff}} < 3$;
- $\simeq 21\%$ lie in a low- ℓ reservoir with $\text{Kn}_{\text{eff}} > 3$.

The reservoir is not permanent. The additional mass needed to reduce Kn_{eff} to unity in all reservoir bins is only

$$M_{\text{req}} \sim 6 \times 10^{-15} \text{ g}. \quad (20)$$

Even if supplied only by the reservoir fraction of the reduced inflow, $\dot{M}_{\text{res}} \sim f_{\text{res}} \chi \dot{M}_B \sim 0.08 \text{ g s}^{-1}$, this mass accumulates in

$$t_{\text{build}} \sim 7 \times 10^{-14} \text{ s}, \quad (21)$$

i.e. on a dynamical timescale. The low- ℓ reservoir therefore grows until it too becomes collisional. This closes the recycling loop: circularization raises the density, the density lowers the mean free path, and collisionality restores angular-momentum redistribution and hydrodynamic inward transport.

8 Hydro-kinetic source coupling and the corrected accretion rate

The calculations above establish that the collisionless region does not behave as a simple loss cone. Most particles miss direct GR capture, lose orbital energy, circularize, and then become marginally or fully collisional because the pile-up-enhanced density shortens the mean free path. The remaining question is how this kinetic pile-up feeds back on the outer hydrodynamic Bondi flow.

The natural representation of this feedback is conservative: the kinetic region returns fluxes of momentum and energy to the hydrodynamic flow. We therefore couple the two regions through source terms derived from the orbit-averaged kinetic model.

8.1 What the kinetic closure computes

At a given time, the hydrodynamic solution provides the local state near the transition region:

$$\rho(r), \quad T(r), \quad v(r), \quad \dot{M}. \quad (22)$$

Using this state, the kinetic closure samples the angular-momentum distribution of particles entering the collisionless region. For each angular-momentum bin it determines the outcome of the orbit:

- direct GR capture, if $\ell < \ell_{\text{crit}}$;
- reflection and rapid re-thermalization near r_{coll} , if the particle does not enter the collisionless region;
- circularization at $r_{\text{circ}}(\ell) = \ell^2/(GM)$, followed by collisionalization if the pile-up-enhanced Knudsen number is below unity or marginal;
- temporary reservoir formation for the low- ℓ tail, which self-collisionalizes after a small additional mass build-up.

The closure then assigns a deposition radius to each component. Reflected particles deposit momentum and energy near r_{coll} . Circularized and collisionalized particles deposit them near their circularization radii, $r_{\text{circ}}(\ell)$. The result is a pair of radial source profiles: a momentum source $S_p(r)$ and an energy source $S_E(r)$.

Physically, $S_p(r)$ is the back-pressure exerted by the thermalizing pile-up on the inflow, while $S_E(r)$ is the kinetic/orbital energy deposited into the gas by thermalization. These source terms are computed from the orbit-averaged kinetic closure; they are not tuned parameters. For the fiducial $M = 10^{-16} M_{\odot}$ case, the integrated momentum feedback corresponds to

$$\frac{P_{\text{pile}}}{P_0} \simeq 5, \quad (23)$$

where P_0 is the characteristic pressure of the baseline hydrodynamic solution near the deposition region. The associated thermalization power is

$$L_{\text{dep}} = 4\pi \int S_E(r) r^2 dr \simeq 4 \times 10^{17} \text{ erg s}^{-1}. \quad (24)$$

8.2 How the sources enter the hydrodynamic equations

The source terms are added directly to the one-dimensional hydrodynamic equations. We do not add a separate net mass source: the material represented by the kinetic closure is recycled material drawn from the same supplied hydrodynamic flux. Its leading effect on the outer one-dimensional flow is therefore the momentum and energy deposited during thermalization/collisionalization.

With the sign convention that the inflow velocity is negative ($v < 0$), a positive S_p opposes the infall:

$$\left. \frac{\partial(\rho v)}{\partial t} \right|_{\text{kin}} = S_p(r), \quad S_p > 0 \quad \text{for outward/back-pressure feedback.} \quad (25)$$

The thermalization energy is added to the gas internal-energy equation:

$$\left. \frac{\partial e}{\partial t} \right|_{\text{kin}} = S_E(r), \quad (26)$$

where e is the gas internal-energy density. Here S_E represents kinetic/orbital energy deposited into the gas by thermalization. Radiative bremsstrahlung losses are still handled separately by the cooling term in the energy equation.

In this formulation the kinetic model supplies the momentum and energy fluxes associated with the pile-up, and the hydrodynamic flow responds self-consistently; no inner temperature or pressure is prescribed by hand.

8.3 Frozen-closure response

As a first test, we compute $S_p(r)$ and $S_E(r)$ from the baseline hydrodynamic profile and then hold them fixed while evolving the flow. We multiply the sources by a dimensionless strength parameter $\alpha \in [0, 1]$,

$$S_p(r) \rightarrow \alpha S_p(r), \quad S_E(r) \rightarrow \alpha S_E(r), \quad (27)$$

and ramp α smoothly over 10^4 hydrodynamic steps. The parameter α is not physical; it is a numerical diagnostic that lets us measure how the hydrodynamic solution responds as the kinetic feedback is gradually turned on.

Table 1: Frozen-closure response: $\chi = \dot{M}/\dot{M}_B$ vs. source strength α for $M = 10^{-16} M_\odot$ ($N = 1200$, order 1). The source profiles are computed from the baseline hydro solution and then held fixed.

α	$\chi = \dot{M}/\dot{M}_B$
0.00	1.000
0.10	0.959
0.25	0.900
0.50	0.811
0.75	0.735
1.00	0.677

The response is smooth and monotonic. No blow-up, shock-like runaway, or limit cycle appears. At full source strength, $\alpha = 1$, the frozen-closure calculation gives

$$\chi_{\text{frozen}} \equiv \frac{\dot{M}}{\dot{M}_B} \simeq 0.68, \quad (28)$$

corresponding to a $\simeq 32\%$ reduction relative to the baseline Bondi value.

8.4 Self-consistent closure update

The frozen calculation assumes that the kinetic source profiles remain equal to their baseline values even after the hydrodynamic flow adjusts. The final calculation removes this approximation. During the hydro evolution, we periodically recompute the kinetic closure from the current hydro state and update the source terms.

Because the kinetic feedback and the subsonic hydro flow respond to each other, the source terms are updated with under-relaxation. If $S_p^{(n)}$ and $S_E^{(n)}$ are the source profiles currently being used by the hydro solver, and $S_{p,\text{kin}}^{(n)}$ and $S_{E,\text{kin}}^{(n)}$ are the newly computed kinetic source profiles from the current hydro state, we update

$$S_p^{(n+1)} = (1 - \omega)S_p^{(n)} + \omega S_{p,\text{kin}}^{(n)}, \quad (29)$$

$$S_E^{(n+1)} = (1 - \omega)S_E^{(n)} + \omega S_{E,\text{kin}}^{(n)}. \quad (30)$$

Here ω is a dimensionless under-relaxation parameter. The limit $\omega = 1$ would replace the old source profiles immediately with the new kinetic prediction. A small value, $\omega \ll 1$, makes the feedback adjust gradually and prevents artificial overshooting. In the fiducial self-consistent calculation we use

$$\omega = 10^{-3}, \quad (31)$$

and recompute the kinetic closure every 1000 hydrodynamic steps. Thus the closure changes slowly compared with a single hydro timestep, while still allowing the kinetic feedback to follow the secular adjustment of the flow.

Table 2: Convergence of the self-consistent source-term calculation for $M = 10^{-16} M_\odot$. The kinetic source profiles are periodically recomputed from the evolving hydro state and updated with $\omega = 10^{-3}$.

Step	$\chi = \dot{M}/\dot{M}_B$
80 000 (end Phase 1)	1.000
100 000	0.884
120 000	0.749
140 000	0.697
160 000	0.686
200 000	0.686
240 000	0.689
280 000	0.692

The system converges monotonically. The final self-consistent value is

$$\chi_{\text{self}} \equiv \frac{\dot{M}}{\dot{M}_B} \simeq 0.69. \quad (32)$$

This is very close to the frozen-closure result, $\chi_{\text{frozen}} \simeq 0.68$, indicating that the kinetic feedback strength does not change strongly as the accretion rate is reduced by $\sim 30\%$.

8.5 Interpretation

The source-term calculation shows that the kinetic pile-up acts as a moderate hydrodynamic back-pressure on the subsonic Bondi flow. It does not shut off the accretion, and it does not produce the fully collisionless suppression $(c_s/c)^2$. For the fiducial $M = 10^{-16} M_\odot$ case, the corrected accretion rate is instead

$$\dot{M} \simeq 0.7 \dot{M}_B. \quad (33)$$

The precise numerical value should still be checked at higher resolution, since the calculation quoted here uses $N = 1200$ cells and order 1 reconstruction. The robust conclusion is that the correction is moderate, of order tens of percent, rather than catastrophic.

9 Mass dependence

The feedback described above matters only when the collisionless transition lies outside the cooling-induced sonic point,

$$r_{\text{sonic}} < r_{\text{coll}} < r_B. \quad (34)$$

In this ordering, the pile-up and its thermalization region lie in the subsonic part of the flow, so the pressure perturbation can communicate upstream and modify the Bondi eigenvalue. For larger PBH masses, r_{sonic} moves outward and overtakes r_{coll} ; the pile-up then lies in the supersonic region and cannot affect the upstream hydrodynamic supply.

Applying the self-consistent source-term closure across the lightest masses gives

$\log_{10}(M/M_\odot)$	$\chi = \dot{M}/\dot{M}_B$	reduction	$r_{\text{coll}}/r_{\text{sonic}}$
-16.0	0.69	31%	8.9
-15.8	0.95	5%	11.5
-15.6	0.94	6%	0.94
-15.3	1.00	0%	0.06

Thus the correction is sharply localized to the lowest masses in the paper. Although $r_{\text{coll}}/r_{\text{sonic}}$ is still large at $\log_{10}(M/M_\odot) = -15.8$, the kinetic source strength and the associated back-pressure are already much weaker than in the $10^{-16} M_\odot$ case, so the correction falls to only a few percent. For $M \gtrsim 10^{-15.5} M_\odot$, the standard one-dimensional Bondi+cooling calculation is self-consistent and no pile-up correction to $\dot{M}_B$ is required.

10 Anomalous collisionality and a conservative lower bound

The kinetic closure used above includes only Coulomb scattering. The recycling flow at r_{coll} is, however, intrinsically counter-streaming: fresh inflowing material, outbound bouncers, and circularizing matter coexist in the same volume with substantial relative velocities. Such configurations are well known to be unstable to current-driven and Weibel-type plasma microinstabilities, which generate magnetic fields *de novo* from noise on the electron skin depth.

For the fiducial $M = 10^{-16} M_\odot$ case, the local plasma frequency gives a skin depth

$$\delta_e = c/\omega_p \sim 5 \times 10^{-8} \text{ cm}, \quad (35)$$

which is already smaller than $r_{\text{coll}} \sim 2 \times 10^{-7} \text{ cm}$. Self-generated magnetic structures therefore fit comfortably inside the relevant flow scale. Equipartition with the bulk kinetic energy would correspond to $B_{\text{eq}} \sim 4 \times 10^8 \text{ G}$; even at a much smaller saturation level $\epsilon_B \sim 10^{-6}$, the proton Larmor radius $r_L \sim 10^{-9} \text{ cm}$ is already well below r_{coll} . The plasma is then magnetized on every scale that matters, and pitch-angle scattering off the wave field acts as anomalous collisionality, much as in collisionless shocks.

Anomalous scattering can only *add* to the Coulomb collisionality already accounted for in Section 7. Its effects all push χ towards unity:

- the low- ℓ reservoir bin self-collisionalizes faster, on a shorter than free-fall timescale;
- part of the marginal $\text{Kn}_{\text{eff}} \sim 1\text{--}3$ population is moved into the collisional regime;
- angular-momentum redistribution near r_{coll} becomes more efficient, increasing the effective loss-cone refilling rate.

The values $\chi \simeq 0.69$ at $10^{-16} M_\odot$ and the mass scan in Section 9 should therefore be read as a *conservative lower bound*: pure Coulomb collisions give the strongest pile-up back-pressure, while any plasma-microinstability contribution moves the system closer to fully hydrodynamic accretion at $\dot{M}_B$. A quantitative determination of the saturated ϵ_B and of the pitch-angle scattering rate would require a dedicated kinetic (PIC-level) calculation and is beyond the scope of this appendix.

11 Fully collisionless limit

The discussion above assumes that the Bondi radius remains collisional, $r_B \gg \lambda_{\text{mfp}}$. For still lower masses, $M \lesssim 10^{-18} M_\odot$, this fails. The gas is then collisionless already at r_B . In that regime there is no pressure-driven Bondi inflow and no collisional outer reservoir through which missed particles can recycle. Most ambient particles undergo ballistic fly-bys; only the GR loss-cone fraction is captured directly.

Bremsstrahlung losses during close passages may radiatively capture a small subset of particles into a bound mini-reservoir, and such a reservoir could self-collisionalize if enough mass accumulates. This would be a local self-regulation process, but it cannot recreate the missing hydrodynamic Bondi supply. The time-averaged rate is instead limited by ballistic and radiative capture from the ambient collisionless bath,

$$\dot{M} \sim \dot{M}_{\text{GR}} + \dot{M}_{\text{rad.cap.}}, \quad \dot{M}_{\text{GR}} \sim \frac{c_s^2}{c^2} \dot{M}_B. \quad (36)$$

A calculation of $\dot{M}_{\text{rad.cap.}}$ requires a separate treatment of radiative capture from a collisionless Maxwellian bath. This regime is outside the mass range of the present paper.

12 Luminosity and remaining uncertainties

The accretion-rate correction is now reasonably well characterized, but the luminosity correction is less certain. The physical bremsstrahlung emissivity in the inner region is controlled by the nonuniform pile-up density, not by a single global multiplier. The orbit-averaged kinetic calculation gives $\langle \xi \rangle_{\text{rec}} \simeq 12$, but this is a residence-time enhancement rather than a spatially uniform factor. The density is expected to be concentrated near turning and circularization radii.

Thus the local emissivity can be substantially enhanced, since $\varepsilon_{\text{ff}} \propto \rho^2$, while the integrated luminosity is bounded by the orbital energy available along the particle trajectories and by the reduced accretion rate. A cumulative diagnostic,

$$L(< r) = 4\pi \int_{r_{\text{in}}}^r \varepsilon_{\text{ff}}(r') r'^2 dr', \quad (37)$$

computed with a regularized orbit-binned density profile, is the next step needed to turn this into a quantitative luminosity correction. For now, the radiative efficiency in the lightest models should be regarded as robust at the order-of-magnitude level, but not yet at the factor-few level.

13 Summary

The collisionless inner region does not invalidate the hydrodynamic Bondi+cooling calculation over the mass range of the paper. The main points are:

1. The gas is collisional at r_B for the masses considered, so a hydrodynamic Bondi supply is established.
2. The flow becomes collisionless only at small radii, $r_{\text{coll}} \sim 7000 r_S$, where particles retain thermal angular momentum and most miss the GR capture cross-section on a single pass.
3. Bremsstrahlung energy losses limit the residence-time pile-up to $\xi \sim 5\text{--}13$, with the orbit-averaged recycling population giving $\langle \xi \rangle_{\text{rec}} \simeq 12$.
4. Missed particles mainly circularize rather than plunge. The resulting pile-up increases the density and lowers the Coulomb mean free path, making most of the circularization region collisional.
5. The remaining low- ℓ reservoir requires only a tiny additional mass to reach $\text{Kn} \sim 1$ and self-eliminates on a dynamical timescale.
6. A conservative hydro-kinetic source coupling gives $\chi \equiv \dot{M}/\dot{M}_B \simeq 0.7$ for $M = 10^{-16} M_{\odot}$, rising to $\chi \simeq 0.95$ by $10^{-15.8} M_{\odot}$ and $\chi = 1$ for $M \gtrsim 10^{-15.5} M_{\odot}$.
7. Anomalous collisionality from plasma microinstabilities in the counter-streaming recycling flow can only add to the Coulomb collisionality and push χ towards unity, so the values above should be read as a conservative lower bound.

8. The fully collisionless $(c_s/c)^2$ suppression applies only once r_B itself is collisionless, below the mass range considered here.

In short, the PBH is not fed by direct loss-cone capture. It is fed by a self-regulated recycling/circularization/collisionalization flow. The main correction is a moderate pressure-feedback reduction of the Bondi supply at the lowest masses, not a catastrophic collisionless suppression.

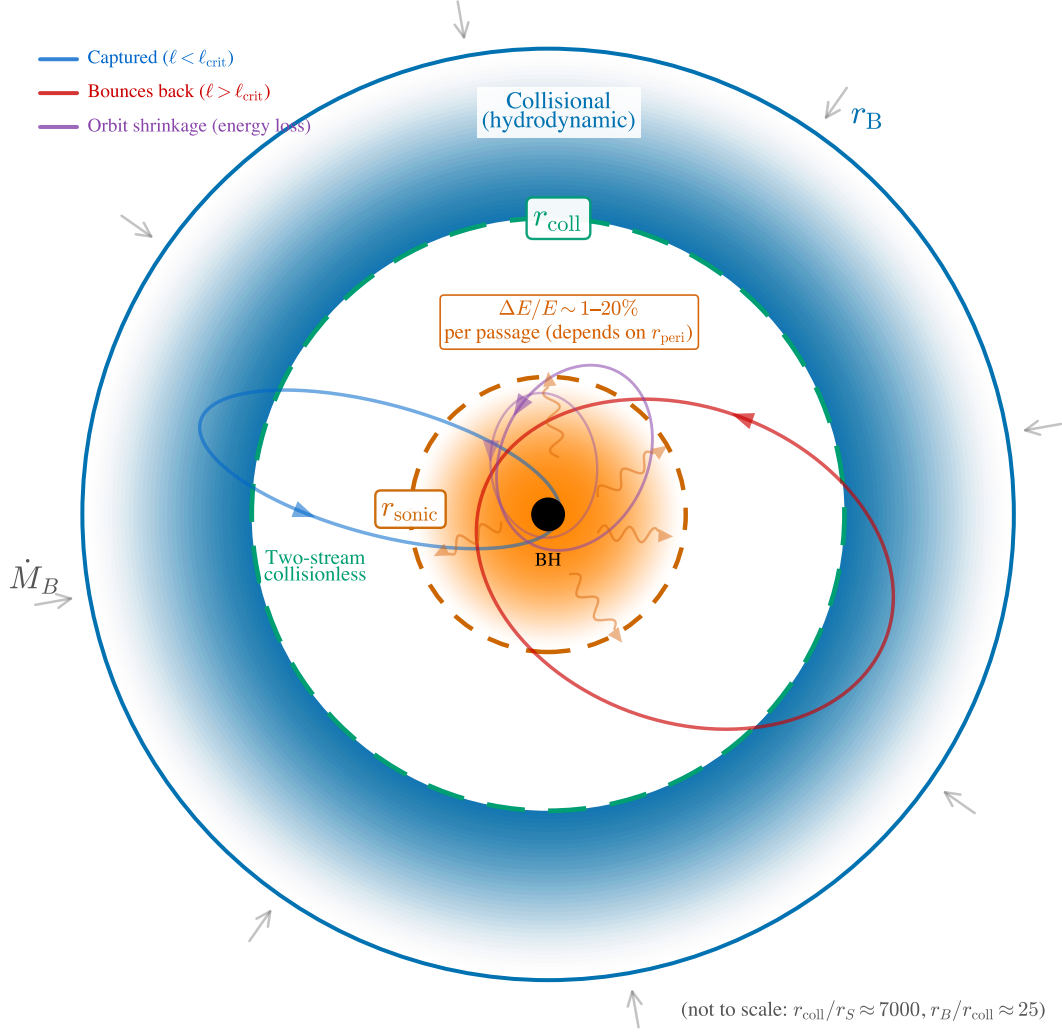


Figure 1: Schematic of the collisionless pile-up and recycling process (not to scale). The flow is hydrodynamic near r_B , becomes collisionless below r_{coll} , and experiences cooling-induced orbit shrinkage near the inner region. Direct GR capture occurs only for the low-angular-momentum loss cone, $\ell < \ell_{\text{crit}}$; most particles miss on the first pass and either return to the collisional zone or circularize.

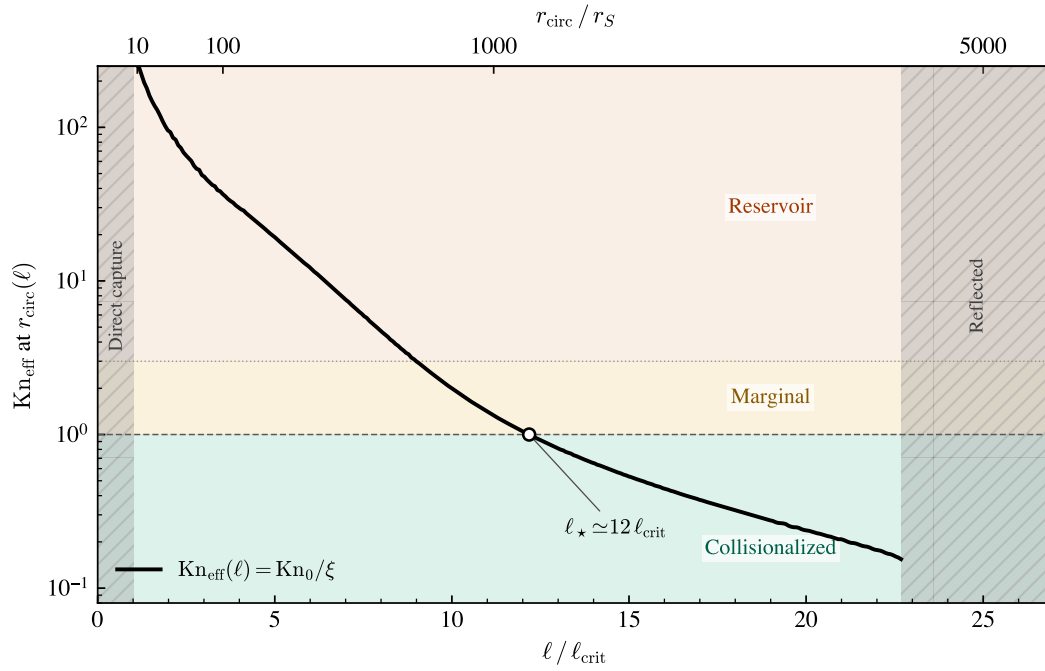


Figure 2: Effective Knudsen number at the circularization radius as a function of angular momentum. For $\ell \gtrsim 12\ell_{\text{crit}}$, the pile-up-enhanced density makes $\text{Kn}_{\text{eff}} < 1$, so the circularization region is collisional. Only the low- ℓ tail remains clearly collisionless. The dashed curve gives r_{circ}/r_S on the right axis.